

# Gina Sohn

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## RESEARCH INTERESTS

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HW/SW Co-design, Compiler, Domain-specific Language, Hardware Accelerator

## EDUCATION

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### Stanford University

Sep. 2022 – Present

*Ph.D., Electrical Engineering*

Advisor: **Kunle Olukotun**

### Seoul National University

Mar. 2018 – Aug. 2022

*Bachelor of Science, Computer Science and Engineering*

Total GPA of 4.16 / 4.3 (Summa Cum Laude)

## PUBLICATIONS

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### Streaming Tensor Program: A streaming abstraction for dynamic parallelism

[Gina Sohn](#), Genghan Zhang, Konstantin Hossfeld, Jungwoo Kim, Nathan Sobotka, Nathan Zhang, Olivia Hsu, Kunle Olukotun

*Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Mar. 2026* [\[paper\]](#)

### Sieve: Dynamic Expert-Aware PIM Acceleration for Evolving Mixture-of-Experts Models

Jungwoo Kim, Rubens Lacouture, Genghan Zhang, [Gina Sohn](#), Qizheng Zhang, Swapnil Gandhi, Christos Kozyrakis, Kunle Olukotun

*arXiv preprint, May 2026 (Under review)* [\[paper\]](#)

### The Dataflow Abstract Machine Simulator Framework

Nathan Zhang, Rubens Lacouture, [Gina Sohn](#), Paul Mure, Qizheng Zhang, Fredrik Kjolstad, Kunle Olukotun

*Proceedings of the International Symposium on Computer Architecture (ISCA), June 2024* [\[paper\]](#)

### 🏆 ISCA Distinguished Artifact Award

### ARK: Fully Homomorphic Encryption Accelerator with Runtime Data Generation and Inter-Operation Key Reuse

Jongmin Kim, Gwangho Lee, Sangpyo Kim, [Gina Sohn](#), John Kim, Minsoo Rhu, and Jung Ho Ahn

*Proceedings of the 55th IEEE/ACM International Symposium on Microarchitecture (MICRO), Oct. 2022* [\[paper\]](#)

### Behemoth: A Flash-centric Training Accelerator for Extreme-scale DNNs

Shine Kim, Yunho Jin, [Gina Sohn](#), Jonghyun Bae, Tae Jun Ham, and Jae W. Lee

*Proceedings of the 19th USENIX Conference on File and Storage Technologies (FAST), Feb. 2021* [\[paper\]](#) [\[slides\]](#)

## WORKSHOP

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### Dynamic Memory for Distributed Architectures

[Gina Sohn](#), Genghan Zhang, Suguna Velury, Nathan Zhang, Kunle Olukotun

*The Sixth Young Architect Workshop (Organized in conjunction with ASPLOS 2024)* [\[pdf\]](#)

## HONORS AND SCHOLARSHIPS

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Graduate Research Fellowship Program, National Science Foundation

2025–2028

Stanford Graduate Fellowship (Burt & Deedee McMurtry Fellow), Stanford University

2022–2025

National Scholarship for Science and Engineering, Korea Student Aid Foundation

2020–2021

Eminence Scholarship, Seoul National University

2019

## RESEARCH EXPERIENCE

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### **Pervasive Parallelism Laboratory**

*Stanford University*

Advisor: Kunle Olukotun

Apr. 2023 – Present

Sep. 2022 – Dec. 2022

### **Stanford Accelerate Laboratory**

*Stanford University*

Advisor: Priyanka Raina

Jan. 2023 – Mar. 2023

### **SCALable Computer Architecture Laboratory**

*Graduate School of Convergence Science and Technology, Seoul National University*

Advisor: Jung Ho Ahn

Jan. 2022 – Apr. 2022

### **Architecture and Code Optimization Laboratory**

*Samsung-SNU Cooperation R&D Center, Seoul National University*

Advisor: Jae W. Lee

Jul. 2020 – Sep. 2020

## TEACHING EXPERIENCE

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### **Digital Systems Design Lab, Stanford University**

*Teaching Assistant*

Apr. 2024 – Jun. 2021

- Topics covered: Software / hardware design tradeoffs, Algorithm design for pipelining and parallelism, System latency and throughput tradeoffs, FPGA optimization techniques.

### **Computer Programming, Seoul National University**

*Teaching Assistant*

Sep. 2021 – Dec. 2021

- Covered C++ and Java programming, Object-Oriented Programming
- Revised lab materials and assignment descriptions, Participated in weekly Professor-TA meetings, Administered laboratories for 170 students every week, Answered questions online

## EXTRACURRICULAR & LEADERSHIP ACTIVITY

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### **Stanford Women in Electrical Engineering (WEE)**

Sep. 2023 – Present

### **Seoul National University College of Engineering Honor Society (STEM)**

*Academic Activity Director (Sep. 2021 – Feb. 2022)*

Mar. 2021 – Jun. 2022

### **Korean-American Women in Science and Engineering (KWise)**

*2021 Mentee Team Leader (Chapter: Los Angeles, USA)*

Mar. 2021 – Dec. 2022

### **Seoul National University Security Research Group (Guardian)**

Mar. 2019 – Jul. 2019